

Project Summary

End-to-End Business Analytics: Retail Sales

Objective

Take raw retail sales data through the full analytics process: clean it in Excel, query it in SQL, model it in Power BI, and turn the results into recommendations for management.

Project contents

File	What it holds
Excel/Raw_Data.xlsx	The original dataset, 12,575 rows, unchanged.
Excel/Data_Analysis.xlsx	Cleaned data, formula analysis, four pivot tables with charts and a business report.
SQL/Business_Analysis.sql	Queries for sales, products, customers and regions, with joins, grouping and filtering.
SQL/retail_business.db	The SQLite database the queries run against.
PowerBI/Business_Intelligence_Dashboard.pbix	Data model, DAX measures and the one-page executive dashboard.
PowerBI/Norysca_DataAnalytics_Task5_SourceData.csv	The cleaned source data loaded into Power BI.
Report/Final_Executive_Report.pdf	Executive summary, findings, dashboard and five recommendations.

Key results

Total sales were €1,552,071 from 11,971 orders, at €129.65 per order, across 25 customers. Sales fell 3.7% in 2023 and rose 6.8% in 2024. Categories differ by 13.5% and regions by 3.1%. January is the peak month. Discounted orders are no larger than full-price orders.

Recommendations in brief

Move customers to higher-priced items, lift order value in Milk Products and Patisserie, plan around the January peak, test whether discounts work, and make item ID a required field at order entry.

How to open the files

Download the files, because GitHub does not preview Excel, Power BI or database files. Open the Excel files in Excel and the .pbix file in Power BI Desktop. For SQL, open retail_business.db in DB Browser for SQLite, then run Business_Analysis.sql in the Execute SQL tab.

Tools

Microsoft Excel, SQLite, Power BI Desktop.